MILITARY SPECIFICATION

RECEIVING SET, COUNTERMEASURES AN/ALR-8

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment specified herein shall be used in military aircraft to receive radio and radar transmissions in the frequency range from 50 to 10750 mc. It shall provide outputs for visual and aural presentation of the signals for analyzing purposes and shall display a band of frequencies on a cathode ray tube such that the frequency of signals within the passband, with respect to the frequency to which the receiver is tuned, may be determined readily.

1.2 Classification - The equipment covered by this specification shall consist of the following:

Item	Type Designation	Applicable Paragraphs
Tuner, Radio Frequency	TN-200/APR-13	3.4.1
Tuner, Radio Frequency	TN-178/APR-13	3.4.2
Tuner, Radio Frequency	TN-179/APR-13	3.4.3
Tuner, Radio Frequency	TN-180/APR-13	3.4.4
Tuner, Radio Frequency	TN-181/APR-13	3.4.5
Mounting	MT-606A/APR-9	3.4.5.2
Radio Frequency Tuner	TN-128B/APR-9	3.4.6
Radio Frequency Tuner	TN-129D/APR-9	3.4.7
Radio Frequency Tuner	TN-130B/APR-9	3.4.8
Radio Frequency Tuner	TN-131C/APR-9	3.4.9
Mixer - Amplifier	CV-124/APR-13	3.4.10
Mixer - Amplifier	CV-43C/APR-9	3.4.11
Control, Radar Set	C-654A/APR-9B	3.4.12
Indicator, Panoramic	ID-226B/APR-9	3.4.13
Mounting	MT-607A/APR-9	3.4.13.1
Power Supply	PP-2848/APR-9	3.4.14
Mounting	MT-605A/APR-9	3.4.14.2
Switch Assembly	SA-418A/ALR-8	3.4.15
Mounting	MT-1512/ALR-8	3.4.15.2

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of

FSC 5895

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MS25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to-failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables and switch assembly, shall not be greater than 192 pounds when carrying two mixer-amplifiers, and four tuning heads. Total weight shall include all major units and accessories (less cables) necessary to make such an equipment completely operative, but shall not include the primary power supply normally common to the aircraft.

3.2.3 Operating Life -

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without the necessity for readjustment of any controls that are inaccessible during flight.

3.2.3.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring replacement or servicing at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.3.3 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.4 Reliability in Mean Time Between Failures - The equipment shall have a minimum of 50 hours of mean (operating) time between failure when tested and accepted as outlined under the requirements of 4.4.2.

3.2.4 Cabling and Connections -

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with Standard MS25245.

3.2.5 Control Panels - All rack or console-mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity before preproduction testing.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0 volt ac and/or 27.5 ±0.5 volt dc

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.11.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of Standard MIL-STD-704, and shall give specified performance from the following power sources with characteristics as defined in Standard MIL-STD-704 having limits as specified herein. The power required shall not be greater than the specified amounts:

(1) A-C power: (single-phase) 115 volt, Category B, 575 volt-ampere

(2) D-C power: 28 volt, Category B, 5.0 amperes

3.2.12.1 Auxiliary A-C Power of the 575 Volt Ampere - Seventy-five (75) volt-amperes of auxiliary a-c power shall be provided for synchros and panoramic sweep motors. Provision shall be made for supplying this power from the main power input or the auxiliary power input by a switch on the front panel of the power supply.

3.2.12.2 Fusing - A 0.25-ampere, slow-burning (SB) fuse shall be provided in the PP-2848/APR-9 power supply for protection of the remote tuning indicator syncro circuit, a 1-ampere fuse for the protection of the -1610-volt circuit, and a 2-ampere fuse for the protection of the remaining combined circuits. One spare fuse in a fuse holder shall be provided for each of the three fuses. The fuse cartridges shall satisfy the requirements of Specification MIL-E-5400.

3.2.12.3 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns to within limits.

3.2.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Warm-Up - When the equipment is initially placed in operation at a subfreezing temperature, a sufficient warm-up period shall be allowed to melt any ice present on moving joints, cams, bearings, etc, before normal operation is expected.

3.3.2 Frequency Characteristics - The complete equipment with nine r-f heads shall cover a frequency range from 50 to 10750 mcs. Each r-f unit shall be calibrated in frequency for direct reading. The calibration accuracy shall be maintained within ±1 percent of the actual frequency throughout any of the service conditions imposed by paragraph 3.2.11. The calibration accuracy at an ambient temperature of 20°C, shall be as specified for the individual tuners. Tuning units shall cover the following ranges:

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the currentversion before use.

Tuner	Frequency (mc)
TN-200/APR-13	50 to 100 mc
TN-178/APR-13	90 to 180 mc
TN-179/APR-13	160 to 320 mc
TN-180/APR-13	300 to 600 mc
TN-181/APR-13	550 to 1100 mc
TN-128B/APR-9	1000 to 2600 mc
TN-129D/APR-9	2300 to 4450 mc
TN-130B/APR-9	4300 to 7350 mc
TN-131C/APR-9	7050 to 10750 mc

3.3.3 Sensitivity -

3.3.3.1 Sensitivity over Frequency Range from 50 to 100 mc (with combination 36.7 and 60-mc i-f amplifier included) - With the first i-f second detector no-signal rectified current adjusted to 0.15 ±0.05 ma, a c-w signal input no greater than -80 dbm into the r-f input panel receptacle from a 50-ohm source shall increase the rectified detector current by a factor of 1.41. This specification shall apply at the maximum response point for any given frequency setting within the tuning range of any r-f tuner under all service conditions imposed by paragraph 3.2.11. A c-w signal no greater than 10 db below the listed value will produce a visible signal on the panoramic indicator.

3.3.3.2 Sensitivity over Frequency Range from 90 to 320 mc (with 60-mc i-f amplifier included) - With the first i-f second detector no-signal rectified current adjusted to 0.15 ±0.05 ma, a c-w signal input no greater than -80 dbm into the r-f input panel receptacle from a 50-ohm source shall increase the rectified detector current by a factor of 1.41. This specification shall apply at the maximum response point for any given frequency setting within the tuning range of any r-f tuner under all service conditions imposed by paragraph 3.2.11. A c-w signal no greater than 10 db below the listed value will produce a visible signal on the panoramic indicator.

3.3.3.3 Sensitivity over Frequency Range from 300 to 1100 mc (with 160-mc i-f amplifier included) - With the first i-f second detector no-signal rectifier current adjusted not to exceed 0.15 ma (tuner connected) and the rectified current not less than 3 times that with tuner disconnected, a c-w signal input no greater than -85 dbm into the r-f input panel receptacle from a 50-ohm source shall increase the rectified detector current by a factor of 1.41. This Specification shall apply at the maximum response point for any given frequency setting within the tuning range of any r-f tuner under all service conditions imposed by paragraph 3.2.11. A c-w signal no more than 10 db below the listed value will produce a visible signal on the panoramic indicator.

3.3.3.4 Sensitivity over Frequency Range from 1000 to 10750 mc (with 160-mc i-f amplifier included) - With standard i-f attenuation control setting (i.e., first i-f second detector no-signal rectified current adjusted not to exceed 0.15 ma (tuner connected) and the rectified current not less than 3 times that with tuner disconnected, a c-w signal input no greater than that specified below into the r-f input receptacle from a 50-ohm source shall produce a signal-plus-noise level in the i-f amplifier which is 3 db greater than the no-signal noise level. The measurement to calibrate the second detector accurately shall be made by using a 93-ohm, 3-db, attenuator between the i-f preamplifier and the main i-f amplifier.

Tuner	Average (dbm) (normal condition)	Minimum (dbm) (normal conditions)	Minimum (dbm) (serv conditions)
TN-128B	-83	-80	-75
TN-129D	-80	-76	-72
TN-130B	-83	-79	-75
TN-131C	-79	-75	-71

3.3.4 Selectivity - The overall selectivity of the system shall be determined by the bandpass characteristics of the tuners and i-f amplifiers, which shall satisfy the requirements of paragraphs 3.3.7 and 3.4.10.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.3.4.1 Selectivity from 1000 to 10750 mc (with the 160-mc i-f amplifier included) - Over the complete tuning range of the respective r-f tuners, the preselector shall be tracked with respect to the oscillator so that its passband is generally centered with the passband of the intermediate frequency. The bandwidth to the extreme 6-db points of the response curve shall be at least 20 mc, using maximum gain as the reference level. In general, gain variation within the passband shall not exceed 6 db; however, the limit of this variation shall be that resulting from the combination of tuners, having respective passband characteristics satisfying the requirements of 3.3.7, with mixer amplifiers satisfying the requirements of 3.4.11.3.1.

3.3.5 Preselectors - Each r-f preselector for the 50- to 1100-mc range shall be a tunable bandpass filter consisting of two lumped-constant, gang-tuned, circuits. Means shall be provided for trimming the tuning of the individual circuits as required to maintain proper tracking of the circuits with each other. Bandpass characteristics of the preselectors shall be such as to satisfy the requirements of paragraph 3.3.7.

3.3.5.1 Preselectors, 50-100 mc, 90-180 mc, and 160-320 mc (TN-200, TN-178, and TN-179) - Preselectors for these frequency ranges shall be tuned by means of variable capacitors having plates shaped to give the circuits linear frequency versus shaft rotation characteristics.

3.3.5.2 Preselectors, 300-600 mc and 550-1100 mc (TN-180 and TN-181) - Preselectors for these tuning ranges shall employ low-inductance tuning capacitors, which shall be ganged and tuned by a single circular-type cam that shall rotate with the local oscillator cam and shall be provided with multiple adjustments, as required, to suitably modify the cam shape to maintain correct preselector tracking with respect to the local oscillator tuning over the tuning range.

3.3.5.3 Preselectors, 1000 to 10750 mc - Each r-f preselector shall be a tunable r-f bandpass filter consisting of three over-coupled, gang-tuned coaxial cavities. Means shall be provided for trimming the tuning of the individual cavities at a number of points throughout the band as required to maintain proper tracking of the cavities with each other. The preselector tuning as a whole shall be tracked with the local oscillator tuning by means of a circular-type cam, which shall rotate with the oscillator cam and shall be provided with multiple adjustments as required to suitably modify the shape to maintain correct preselector tracking with respect to local oscillator tuning over the tuning range. Bandpass characteristics of the preselector shall satisfy the requirements of 3.3.7.

3.3.5.4 I-F Preamplifier, 160 mc - The i-f preamplifier included in each tuning unit shall provide sufficient amplification to produce, in conjunction with the first i-f amplifier and second detector of the mixer amplifier (paragraph 3.4.11.3.1), a noise output (with tuner connected) of at least 3 times that of tuner disconnected. This specification shall apply with 0 db i-f attenuation and automatic gain control off. The bandwidth of the preamplifier shall be such as to enable the conditions of paragraph 3.3 to be fulfilled.

3.3.6 Tuning Requirements - Each tuning unit shall normally tune remotely from the C-654A/APR-9B remote control with provision for either manual or automatic motor-drive tuning over any chosen sector portion of its frequency range. A clutch shall be provided to disengage the tuning mechanism from the motor drive to make manual tuning possible to the r-f tuner front panel by means of the local tuning control. Under conditions of automatic motor-drive tuning (selector switch on the C-654A/APR-9B at its SECTOR-SWEEP setting), the limits of frequency being swept shall be adjustable from the remote control over any chosen frequency interval down to a minimum interval of not greater than that specified under "Detailed Requirements" (paragraph 3.4) for the particular tuner. In addition, with the unit tuned to any frequency in its range after a period of MANUAL operation, setting of the selector switch to its SECTOR-SWEEP position shall cause the unit to return to the indicated sector-sweep frequency range and operate properly between the indicated sweep limits.

3.3.7 Tuner Passband - The passband of the individual tuners, including the combination of preselector and preamplifier, shall satisfy the following conditions:

3.3.7.1 50 to 1000 mc -

The bandpass variation of the individual tuners shall be measured by maintianing an r-f power input at constant frequency and at such a level as to maintain a constant output from the tuner preamplifier of approximately 50 mv into a 90-ohm matched load as the tuner frequency is varied so as to provide at the output all intermediate frequency from one bandpass limit to the other (i.e., 34.7 to 38.7 mc for the TN-200, 57.5 to 62.5 mc for the TN-178 and TN-179, 155 to 165 mc for the TN-180 and TN-181).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

This measurement shall be made at a minimum of five bandpass sectors equally spaced through the frequency range of the tuner. At no more than one such sector shall the variation exceed 6 db. At no more than two such sectors shall the variation exceed 4.5 db, including the abovementioned sector. The average of the variation measured in all sectors shall be no greater than 4 db.

3.3.7.2 1000 to 10750 mc -

(1) When the r-f power input to a tuner is maintained at constant frequency and at such a level as to maintain a constant output from the tuner preamplifier of approximately 50 mv into a 90-ohm matched load -

(a) The maximum variation in input r-f power level shall not exceed 6 db of this value as the tuner frequency is varied so as to produce at the output all intermediate frequencies from 150 to 170 mc;

(b) variation in the passband shall not be recorded at more than one point when measurements are made in the same manner at different points throughout the frequency band of the tuner;

(c) an increase in input power level of 4.5 db above the minimum required shall not be recorded in the passband at more than two out of a minimum of five 20-mc sectors, evenly spaced throughout the frequency band of the tuner, and the average increase in power required from the minimum to produce output in the same 20-mc bands measured above shall not exceed 4 db.

(2) Image rejection of the various tuners shall satisfy the following requirements:

(a) TN-128B/APR-9 - The image frequency for this tuner is 320 mc higher than the the indicated frequency, and the rejection to a signal at the image frequency shall be at least 65 db at any part of the tuning range.

(b) TN-129D/APR-9 - The image frequency of this tuner is 320 mc lower than the indicated frequency, and the rejection to a signal at the image frequency shall be at least 60 db at any part of the tuning range.

(c) TN-130B/APR-9 - The image frequency for this tuner is 320 mc higher in frequency than the desired signal, and the rejection at the image frequency shall be at least 55 db at any part of the tuning range.

(d) TN-131C/APR-9 - The image frequency for this tuner is 320 mc lower in frequency than the desired signal. Rejection to a signal at the image frequency, in general, shall be at least 55 db at any frequency throughout the tuning range of the tuner. If isolated points are found where the rejection goes below this figure, tuners having such points of lower rejection will be acceptable provided the image rejection at such points is at least 50 db and provided no more than two such points can be found in the range of the tuners, and the combined width in frequency of the areas wherein the rejection is lower than 55 db shall be not more than 15 percent of the frequency range of the tuner.

3.3.8 Spurious Response Rejection - (50 to 1100 mc) - All spurious responses to a given signal and to signals at the image frequency shall be at least 60 db below the desired response in all r-f tuners. The i-f signal rejection shall be at least 55 db below the desired response for all r-f tuners.

3.3.9 Overall Pulse Response (Panoramic) - The overall pulse response of the receiver shall be such that the amplitude of pulse signals as measured on the face of the ID-226B/APR-9 indicator cathode-ray tube shall not vary by more than a 2-to-1 ratio as the pulsewidth of an r-f input signal is reduced from 10 to 3 us with the CV-124/APR-13 mixer-amplifier connected, and from 10 to 1.5 usec with the CV-43C/APR-9 connected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.4 Detail Requirements -

3.4.1 R-F Tuner, TN-200/APR-13 (50 - 100 mc) - This r-f tuner shall fulfill the following requirements in addition to the applicable requirements of 3.3.2.

3.4.1.1 Function - This unit shall perform a function of preselection and conversion of radio frequencies within the range from 50 to 100 mc to an intermediate frequency of 36.7 mc.

3.4.1.2 Form Factor and Mounting - The unit shall be housed in a size A1-D case and mounted with a size MT-A1-D vibration mount, MT-606A/APA-9, in accordance with Specification MIL-C-172.

3.4.1.3 Contents of Unit - This unit shall contain the following items:

(1) Local oscillator and crystal mixer

(2) I-F preamplifier

(3) Frequency indicator counter

(4) R-F bandpass filter

(5) Two circuit r-f preselectors

(6) Filament transformer for local oscillator and i-f preamplifier

(7) Sector-sweep and control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism and braking circuit

3.4.1.3.1 Local Oscillator and Crystal Mixer - The local oscillator shall employ-lumped circuit parameters. Its frequency shall be greater than the r-f preselector resonant frequency (the indicated receiver frequency) by the amount of the first i-f amplifier frequency (36.7 mc). The mixer shall use a crystal diode. The local oscillator shall supply adequate injection to the mixer for optimum sensitivity throughout the entire tuning range of the r-f tuner, and it shall be suitably decoupled from the signal circuits. The meter on the panel of the receiver control unit shall indicate by an upward deflection that the oscillator is functioning. The meter shall read zero when the oscillator is disabled, so that there is no oscillator injection.

3.4.1.3.2 I-F Preamplifier - The i-f preamplifier shall have a power gain of 40 db at the frequency of maximum response in the bandpass. The preamplifier bandwidth shall be such as to enable the conditions of paragraph 3.3.7 to be fulfilled.

3.4.1.3.3 Frequency Indicator Counter - The frequency indicator counter shall be provided on the front panel for maintenance purposes, checking of remote frequency indication, and for bench servicing. This frequency counter shall indicate a received signal freuqency to an accuracy of at least ±0.5 mc within the tuning range at an ambient temperature of 20° C.

3.4.1.4 Weight with Mounting - The weight of this unit shall not be greater than 21 pounds, with the mounting base included.

3.4.1.5 Electrical Connections - All electrical connections shall be made to the front panel and shall consist of the following:

	Connecting Plug
Power and Control Input Receptacle	P-101
R. F. Input Receptacle	P-102
I-F Preamplifier Output Receptacle	P-103

Receptacles shall be marked externally on the front panel with the connection plug number.

Source: https://assist.dla.mil -- Downloaded: 20260-19T15:01Z
Check the source to verify that this is the currentversion before use.

3.4.1.6 Controls - The unit shall contain no controls requiring adjustment during operation of the equipment.

3.4.1.7 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall, in addition, simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 50 + 0.1 - 0. At the high-frequency end of the tuning range, such adjustment shall be possible at the indicated r-f tuner frequency ±0.2 mc.

3.4.1.8 Tuning Requirements - Under conditions of automatic motor-drive tuning, (SECTOR SWEEP - MAN switch on the C-654A/APR-9 control at its SECTOR-SWEEP setting,) the limits of frequency being swept shall be adjustable from the receiver control over any chosen frequency down to a minimum interval not greater than 5 mc. The tuning rate condition of SECTOR-SWEEP operation shall be such that the time required to cover the complete frequency of the r-f tuner is between the limits of 20 ±5 seconds, at 20° C and is less than 50 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector sweep limits, and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B control. After removal of tuning motor excitation, the receiver tuning motion shall stop with a maximum interval of 0.1 mc at any frequency in the range of the r-f tuner.

3.4.2 R-F Tuner TN-178/APR-13 (90 to 180 mc) - This r-f tuner shall fulfill the following requirements in addition to the applicable requirements of paragraph 3.3.

3.4.2.1 Function - This unit shall perform the function of preselection and conversion of radio frequencies within the range from 90 to 180 mc to an intermediate frequency of 60 mc.

3.4.2.2 Form Factor and Mounting - This unit shall be housed in a size A1-D case and mounted with a size MT-A1-D mounting, MT-606A/APR-9, fitted with vibration isolators satisfying the requirements of Specification MIL-C-172.

3.4.2.3 Contents of Unit - This unit shall contain the following major items:

(1) Local oscillator and crystal mixer

(2) I-F preamplifier

(3) Frequency indicator counter

(4) R-F bandpass filter

(5) Two circuit r-f preselectors

(6) Filament transformer for local oscillator and i-f preamplifier

(7) Sector-sweep and control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism and braking circuit

3.4.2.3.1 Local Oscillator and Crystal Mixer - The local oscillator shall have lumped circuit parameters. Its frequency shall be greater than the r-f preselector resonant frequency (the indicated receiver frequency) by the amount of the first i-f amplifier frequency (60 mc). The mixer shall use a crystal diode. The local oscillator shall supply adequate injection to the mixer for optimum sensitivity throughout the entire tuning range of the r-f tuner, and it shall be suitable decoupled from the signal circuits. The meter on the panel of the receiver control unit shall indicate an upward deflection that the oscillator is functioning and shall read zero when the oscillator is disabled so that there is no oscillator injection.

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.2.3.2 I-F Preamplifier - The i-f preamplifier shall have a power gain of 30 db at the frequency of maximum response in the bandpass. The preamplifier bandwidth shall be such as to enable the conditions of paragraph 3.3.7 to be fulfilled.

3.4.2.3.3 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for maintenance purposes, for checking of remote frequency indication, and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±1.0 mc within the tuning range at an ambient temperature of 20° C.

3.4.2.4 Weight with Mounting - The weight of this unit shall not be greater than 20 pounds, with the mounting base included.

3.4.2.5 Electrical Connections - All electrical connections shall be made to the front panel and shall consist of the following:

	Connecting Plug
Power and control input receptacle	P-101
R-F input receptacle	P-102
I-F preamplifier output receptacle	P-103

Receptacles shall be marked externally on the front panel with the connecting plug number.

3.4.2.6 Controls - This unit shall contain no controls requiring adjustment during operation of the equipment.

3.4.2.7 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning-range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall, in addition, simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 90 + 0.2/-0 mc. At the high-frequency end of the tuning range, such adjustment shall be possible at the indicated r-f tuner frequency ±0.4 mc.

3.4.2.8 Tuning Requirements - Under conditions of automatic motor drive tuning (SECTOR-SWEEP-MAN switch on the C-654A/APR-9B control at its SECTOR-SWEEP setting) the limits of frequency being swept shall be adjustable from the receiver control over any chosen frequency down to a minimum interval not greater than 10 mc. The tuning rate under conditions of SECTOR SWEEP operation shall be such that the time required to cover the complete frequency of the r-f tuner is between the limits of 20 ±5 seconds at 20° C and is less than 50 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector sweep limits, and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B receiver control. After removal of tuning motor excitation, the receiver tuning motion shall stop with a maximum interval of 0.2 mc at any frequency in the range of the r-f tuner.

3.4.3 R-F Tuner TN-179/APR-13 (160-320 mc) - This r-f tuner shall fulfill the following requirements in addition to the applicable requirements of paragraph 3.3.

3.4.3.1 Function - This unit shall perform the function of preselection and conversion of radio frequencies within the range from 160 to 320 mc to an intermediate frequency of 60 mc.

3.4.3.2 Form Factor and Mounting - This unit shall be housed in a size A1-D case and mounted with a size MT-A1-D mounting base, MT-606A/APR-9, fitted with vibration isolators in accordance with Specification MIL-C-172.

3.4.3.3 Contents of Unit - This unit shall contain the following major items:

(1) Local oscillator and crystal mixer

(2) I-F preamplifier

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

(3) Frequency indicator counter

(4) R-F low pass filter

(5) Two circuit r-f preselectors

(6) Filament transformer for local oscillator and i-f preamplifier

(7) Sector-Sweep and control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism and braking circuit

3.4.3.3.1 Local Oscillator and Crystal Mixer - The local oscillator shall use lumped circuit parameters. Its frequency shall be greater than the r-f preselector resonant frequency (the indicated receiver frequency) by the amount of the first i-f amplifier frequency (60 mc). The mixer shall use a crystal diode. The local oscillator shall supply adequate injection to the mixer for optimum sensitivity throughout the entire tuning range of the r-f Tuner, and it shall be suitable decoupled from the signal circuits. The meter on the panel of the receiver control unit shall indicate by an upward deflection that the oscillator is functioning and shall read zero when the oscillator is disabled so that there is no oscillator injection.

3.4.3.3.2 I-F Preamplifier - The i-f preamplifier shall have a power gain of 30 db at the frequency of maximum response in the bandpass. The preamplifier bandwidth shall be such as to enable the conditions of paragraph 3.3.7 to be fulfilled.

3.4.3.3.3 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for maintenance purposes, for checking of remote frequency indication, and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±1.0 mc within the tuning range at an ambient temperature of 20° C.

3.4.3.4 Weight with Mounting - The weight of this unit shall not be greater than 20 pounds, with the mounting base included.

3.4.3.5 Electrical Connections - All electrical connections shall be made to the front panel and shall consist of the following:

	Connecting Plug
Power and control input receptacle	P-101
R-F input receptacle	P-102
I-F preamplifier output receptacle	P-103

Receptacles shall be marked externally on the front panel with the connecting plug number.

3.4.3.6 Controls - This unit shall contain no controls requiring adjustment during operation of the equipment.

3.4.3.7 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall, in addition, simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 160 + 0.4/-0 mc. At the high-frequency end of the tuning range, such adjustment shall be possible at the indicated r-f tuner frequency ±0.8 mc.

3.4.3.8 Tuning Requirements - Under conditions of automatic motor drive tuning (SECTOR SWEEP - MAN switch on the receiver control at its SECTOR-SWEEP setting,) the limits of frequency being swept shall be adjustable from the receiver control over any chosen frequency down to a minimum increment not greater

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

than 16 mc. The tuning rate under conditions of SECTOR SWEEP operation shall be such that the time required to cover the complete frequency of the r-f Tuner is between the limits of 20 ±5 seconds at 20° C and is less than 50 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector sweep limits, and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B receiver conrtol. After removal of tuning motor excitation, the receiving tuning motion shall stop with a maximum interval of 0.4 mc at any frequency in the range of the r-f tuner.

3.4.4 R-F Tuner TN-180/APR-13 (300 - 600 mc) - This r-f tuner shall fulfill the following requirements in addition to the applicable requirements of paragraph 3.3.

3.4.4.1 Function - This unit shall perform the function of preselection and conversion of radio frequencies within the range from 300 to 600 mc to an intermediate frequency of 160 mc.

3.4.4.2 Form Factor and Mounting - This unit shall be housed in a size A1-D case and mounted with a size MT-A1-D mounting base, MT-606A/APR-9, fitted with vibration isolators in accordance with Specification MIL-C-172.

3.4.4.3 Contents of Unit - This unit shall contain the following major items:

(1) Local oscillator and crystal mixer

(2) I-F preamplifier

(3) Frequency indicator counter

(4) R-F low pass filter

(5) Two circuit r-f preselectors

(6) Filament transformer for local oscillator and i-f preamplifier

(7) Sector-Sweep and control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism and braking circuit

3.4.4.3.1 Local Oscillator and Crystal Mixer - The local oscillator shall use lumped circuit parameters. Its frequency shall be greater than the r-f preselector resonant frequency (indicated receiver frequency) by the amount of the first i-f amplifier frequency (160 mc). The mixer shall use a crystal diode. The local oscillator shall supply adequate injection to the mixer for optimum sensitivity throughout the entire tuning range of the r-f Tuner and it shall be suitably decoupled from the signal circuits. The meter on the panel of the receiver control unit shall indicate whether the injection level is satisfactory.

3.4.4.3.2 I-F Preamplifier - The i-f preamplifier shall have a power gain of 30 db. The preamplifier bandwidth shall be such as to enable the conditions of pargaraph 3.3.7 to be fulfilled.

3.4.4.3.3 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for maintenance purposes, for checking of remote frequency indication, and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±3 mc within the tuning range at an ambient temperature of 20° C.

3.4.4.4 Weight with Mounting - The weight of this unit shall not be greater than 22 pounds, with the mounting base included.

3.4.4.5 Electrical Connections - All electrical connections shall be made to the front panel and shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the current version before use.

	Connecting Plug
Power and control input receptacle	P-101
R-F input receptacle	P-102
I-F preamplifier output receptacle	P-103

Receptacles shall be marked externally on the front panel with the connecting plug number.

3.4.4.6 Controls - This unit shall contain no controls requiring adjustment during operation of the equipment.

3.4.4.7 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall, in addition, simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 300 + 0.6/-0 mc. At the high-frequency end of the tuning range, such adjustment shall be possible at the indicated r-f Tuner frequency ±1.0 mc.

3.4.4.8 Tuning Requirements - Under conditions of automatic motor drive tuning (SECTOR SWEEP - MAN switch on the C-654A/APR-9B control at its SECTOR SWEEP setting) the limits of frequency being swept shall be adjustable from the receiver control over any chosen frequency down to a minimum interval not greater than 30 mc. The tuning rate under condition of SECTOR SWEEP operation shall be such that the time required to cover the complete frequency of the r-f. tuner is between the limits of 20 ±5 seconds at 20° C and is less than 50 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector sweep limits, and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B receiver control. After removal of tuning motor excitation, the receiver tuning motion shall stop with a maximum interval of 0.6 mc at any frequency in the range of the r-f tuner.

3.4.5 R-F Tuner TN-181/APR-13 (550 - 1100 mc) - This r-f tuner shall fulfill the following requirements in addition to the applicable requirements of paragarph 3.3.

3.4.5.1 Function - This unit shall perform the function of preselection and conversion of radio frequencies within the range from 550 to 1100 mc to an intermediate frequency of 160 mc.

3.4.5.2 Form Factor and Mounting - This unit shall be housed in a size A1-D case and mounted with a size MT-A1-D mounting base, MT-606A/APR-9, in accordance with Specification MIL-C-172.

3.4.5.3 Contents of Unit - This unit shall contain the following major items:

(1) Local oscillator and crystal mixer

(2) I-F preamplifier

(3) Frequency indicator counter

(4) R-F low pass filter

(5) Two circuit r-f preselectors

(6) Filament transformer for local oscillator and i-f preamplifier

(7) Voltage divider circuit for local oscillator repeller

(8) Sector-Sweep and control potentiometer

(9) Synchro generator for remote frequency indication

(10) Motor-Drive tuning mechanism and braking circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

(11) Radar blanking input circuit receptacle

(a) Each r-f tuner shall be provided with a connection from the radar blanking input receptacle through an appropriate d-c blocking capacitor to the reflector terminal of the local oscillator tube. The circuit shall permit blanking of the tuner when a pulse voltage in the range from 40 to 125 volts peak-to-peak is applied to the radar blanking input receptacle. The applied blanking pulse is to have a duration of from 6 to 15 us and a repetition rate of from 20 to 20,000 pulses per second. Spurious response resulting from the reflector modulation of the local oscillator reflector shall be at least 60 db below the desired response.

3.4.5.3.1 Local Oscillator and Crystal Mixer - The local oscillator shall be a klystron in a tunable coaxial cavity. Its frequency shall be greater than the r-f preselector resonant frequency (the indicated receiver frequency) by the amount of the first i-f amplifier frequency (160 mc). The mixer shall use a crystal diode. The local oscillator shall supply adequate injection to the mixer for optimum sensitivity throughout the entire tuning range of the r-f tuner, and it shall be suitably decoupled from the signal circuits. The meter on the panel of the receiver control unit shall indicate whether the injection level is satisfactory.

3.4.5.3.2 I-F Preamplifier - The i-f preamplifier shall have a power gain of 30 db. The preamplifier bandwidth shall be such as to enable the conditions of paragraph 3.3.7 to be fulfilled.

3.4.5.3.3 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for maintenance purposes, for checking of remote frequency indication, and for bench servicing. The frequency counter shall indicate a received signal frequency to an accuracy of at least ±5 mc within the tuning range of an ambient temperature of 20° C.

3.4.5.4 Weight with Mounting - The weight of this unit shall not be greater than 23.5 pounds, with the mounting base included.

3.4.5.5 Electrical Connections - All electrical connections shall be made to the front panel and shall consist of the following:

	Connecting Plug
Power and control input receptacle	P-101
R-F input receptacle	P-102
I-F preamplifier output receptacle	P-103
Radar blanking input receptacle	P-3014

Receptacles shall be marked externally on the front panel with the connecting plug number.

3.4.5.6 Controls - This unit shall contain no controls requiring adjustment during operation of the equipment.

3.4.5.7 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall, in addition, simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 550 + 1.0/-0 mc. At the high-frequency end of the tuning range, such adjustment shall be possible at the indicated r-f tuner frequency ±2.0 mc.

3.4.5.8 Tuning Requirements - Under conditions of automatic drive tuning (SECTOR SWEEP-MAN switch on the C-654A/APR-9B control at its SECTOR SWEEP setting) the limits of frequency being swept shall be adjustable from the receiver control over any chosen frequency down to a minimum interval not greater than 55 mc. The tuning rate under condition of SECTOR SWEEP operation shall be such that the time required to cover the complete frequency of the r-f tuner is between the limits of 40 ±5 seconds at 20° C and is less than 70 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector sweep limits, and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B receiver control. After removal of tuning motor excitation, the receiver tuning motion shall stop with a maximum interval of 1.0 at any frequency in the range of the r-f tuner.

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.6 R-F Tuner TN-128/APR-9 (1000 - 2600 mc) - This r-f tuner shall fulfill the following requirements:

3.4.6.1 Form Factor and Mounting - This unit shall be housed in a size A1-D case in accordance with Specification MIL-C-172, and shall be supplied with an MT-A1-D mounting base fitted with vibration isolators.

3.4.6.2 Contents - This unit shall contain the following major items:

(1) Three-Cavity r-f preselector

(2) Reflex-Klystron local oscillator

(3) Crystal mixer

(4) I-F preamplifier

(5) Local oscillator and i-f preamplifier filament transformer

(6) Local oscillator repeller voltage divider circuit

(7) Sector-Sweep control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism

(10) Frequency indicator counter

(11) Radar blanking input circuit receptacle -

(a) Each r-f tuner shall be provided with a connection from the radar blanking input receptacle through an appropriate d-c blocking capacitor to the reflector terminal of the local oscillator tube. The circuit shall permit blanking of the tuner when a pulse voltage in the range from 40 to 125 volts peak-to-peak is applied to the radar blanking input receptacle. The applied blanking pulse is to have a duration of from 6 to 15 us and a repetition rate of from 20 to 20000 pulses per second. Spurious responses resulting from the reflector modulation of the local oscillator reflector shall be at least 60 db below the desired response.

3.4.6.3 Weight with Mounting - The weight of this unit shall not be greater than 22 pounds, with the mounting base included.

3.4.6.4 Electrical Connections - Electrical connections shall be made to the front panel and shall consist of the following:

Power and control input receptacle
R-F input receptacle
I-F preamplifier output receptacle
Radar blanking input receptacle

3.4.6.5 Controls - A local tuning control for bench servicing (with provision for disengaging the motor drive) shall be located on the front panel.

3.4.6.6 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning end of the tuning range shall simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the current version before use.

possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 1000 + 2/-0 mc. At the high-frequency end of the tuning range, such adjustments shall be possible at the indicated r-f tuner frequency of 2600 ±4 mc.

3.4.6.7 Tuning Rate - The tuning rate under condition of SECTOR-SWEEP operation shall be such that the time required to cover the complete frequency of the r-f tuner is between the limits of 29 and 75 seconds at 20° C and in less than 95 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector-sweep limits and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B remote control. After removal of tuning motor excitation, the receiver tuning motion shall stop with a maximum interval of 2 mc at any frequency in the range of the r-f tuner.

3.4.6.8 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for purposes of checking the synchronism of remote frequency indication and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±5 mc within the tuning range at an ambient temperature of 20° C.

3.4.6.9 Local Oscillator and Crystal Mixer - The local oscillator shall be a coaxial cavity type using a reflex klystron tube. Its frequency shall be greater than the r-f preselector resonant frequency (the indicator receiver frequency) by the amount of the first i-f amplifier frequency (160 mc). The mixer shall use a crystal; it shall be parallel fed by the local oscillator and received signals. The input level to the mixer from the local oscillator shall be of such magnitude as to maintain the crystal current, as indicated by the remote control metering circuit, within the range from 0.25 to 6 ma over the complete tuning range of the r-f tuner.

3.4.6.10 Sector-Sweep Adjustment - During the SECTOR-SWEEP operation, it shall be possible to adjust the frequency interval from the C-654A/APR-9B remote control to any desired interval within the tuning range down to a minimum interval of not greater than 150 mc.

3.4.7 R-F Tuner TN-129D/APR-9 (2300 - 4450 mc) - This r-f tuner shall fulfill the following requirements:

3.4.7.1 Form Factor and Mounting - This unit shall be housed in a size A1-D case in accordance with Specification MIL-C-172 and shall be supplied with an MT-A1-D mounting base fitted with vibration isolaters.

3.4.7.2 Contents - This unit shall contain the following major items:

(1) Three-Cavity r-f preselector

(2) Reflex-Klystron local oscillator

(3) Crystal mixer

(4) I-F preamplifier

(5) Local oscillator and I-F preamplifier filament transformer

(6) Local oscillator repeller voltage driver circuit

(7) Sector-Sweep control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism

(10) Frequency indicator counter

(11) Radar blanking input circuit receptacle.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the current version before use.

(a) Each r-f tuner shall be provided with a connection from the radar blanking input receptacle through an appropriate d-c blocking capacitor to the reflector terminal of the local oscillator tube. The circuit shall permit blanking of the tuner when a pulse voltage in the range of from 40 to 125 volts peak-to-peak is applied to the radar blanking input receptacle. The applied blanking pulse is to have a duration of from 6 to 15 usec and a repetition rate of from 20 to 20000 pulses per second. Spurious responses resulting from the reflector modulation of the local oscillator reflector shall be at least 60 db below the desired response.

3.4.7.3 Weight with Mounting - The weight of this unit shall not be greater than 23 pounds with the mounting base included.

3.4.7.4 Electrical Connections - Electrical connections shall be made to the front panel and shall consist of the following:

Power and control input receptacle
R-F input receptacle
I-F preamplifier output receptacle
Radar blanking input receptacle

3.4.7.5 Controls - A local tuning control for bench servicing (with provision for disengaging the motor drive) shall be located on the front panel.

3.4.7.6 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltage from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.3). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 2300 + 2/-0 mc. At the high-frequency end of the tuning range, such adjustments shall be possible at the indicated r-f tuner frequency of 4450 ±5 mc.

3.4.7.7 Tuning Rate - The tuning rate under condition of SECTOR-SWEEP operation shall be such that the time required to cover the complete frequency range of the r-f tuner is between the limits of 29 and 75 seconds at 20° C and in less than 95 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector-sweep limits, and shall be maintained for operation with MANUAL switch setting in the C-654A/APR-9B control. After removal of tuning motor excitation, the receiver tuning motion shall stop within a maximum interval of 2 mc at any frequency in the range of the r-f tuner.

3.4.7.8 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for purposes of checking the synchronism of remote frequency indication and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±5 mc within the tuning range at an ambient temperature of 20° C.

3.4.7.9 Local Oscillator and Crystal Mixer - The local oscillator shall be a coaxial cavity type, using a reflex kylstron tube. Its frequency shall be less than the r-f preselector resonant frequency (the indicated receiver frequency) by the amount of the preamplifier center frequency (160 mc). The mixer shall use a crystal. The input level to the mixer from the local oscillator shall be of such magnitude as to maintain the crystal current, as indicated by the remote control metering circuit, between the limits of 0.25 and 6 ma within the range of the tuner.

3.4.7.10 Sector Sweep Adjustment - During SECTOR-SWEEP operation it shall be possible to adjust the frequency interval from the remote C-654A/APR-9B control to any desired interval within the tuning range down to a minimum interval of not greater than 150 mc.

3.4.8 R-F Tuner TN-130B/APR-9 (4300 - 7350 mc) - This r-f tuner shall fulfill the following requirements:

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.8.1 Form Factor and Mounting - This unit shall be housed in a size A1-D case in accordance with Specification MIL-C-172 and shall be supplied with an MT-A1-D mounting base fitted with vibration isolators.

3.4.8.2 Contents - This unit shall contain the following major items:

(1) Three-Cavity r-f preselector

(2) Reflex-Klystron local oscillator

(3) Crystal mixer

(4) I-F preamplifier

(5) Local oscillator and i-f preamplifier filament transformer

(6) Local oscillator reflector voltage divider circuit and control grid bias circuit

(7) Sector-Sweep control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism

(10) Frequency indicator counter

(11) Radar blanking input circuit receptacle

(a) Each r-f tuner shall be provided with a connection from the radar blanking input receptacle through an appropriate d-c blocking capacitor to the reflector terminal of the local oscillator tube. The circuit shall permit blanking of the tuner when a pulse voltage in the range from 40 to 125 volts peak-to-peak is applied to the radar blanking input receptacle. The applied blanking pulse is to have a duration of from 20 to 20000 pulses per second. Spurious responses resulting from the reflector modulation of the local oscillator reflector shall be at least 60 db below the desired response.

3.4.8.3 Weight with Mounting - The weight of this unit shall not be greater than 24 pounds with the mounting base included.

3.4.8.4 Electrical Connections - Electrical connections shall be made to the front panel and consist of the following:

Power and control input receptacle
R-F input receptacle
I-F preamplifier output receptacle
Radar blanking input receptacle

3.4.8.5 Controls - A local tuning control for bench servicing with provision for disengaging motor drive (paragraph 3.3.6) shall be located on the front panel.

3.4.8.6 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range which will remove the exciting voltages from the tuning motor. Operation of the limit switch at the low frequency end of the tuning range shall simultaneously apply a voltage at the ID-226B/APR-9 indicator which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 4300 ±4/-0 mc. At the high-frequency end of the tuning range, such adjustments shall be possible at the indicated r-f tuner frequency of 7350 ±5 mc.

Source: https://assist.dla.mil -- Downloaded: 20260-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.8.7 Tuning Rate - The tuning rate under condition of SECTOR-SWEEP operation shall be such that the time required to cover the complete frequency range of the r-f tuner is between the limits of 45 and 85 seconds at 20° C and in less than 125 seconds at -55° C. This tuning rate shall be maintained for all possible settings of the sector-sweep limits and shall be maintained for operation with MANUAL switch setting in the remote control C-654A/APR-9B. After removal of tuning motor excitation, the receiver tuning motion shall stop within a maximum interval of 4 mc at any frequency range of the r-f tuner.

3.4.8.8 Indicators - A frequency indicator counter shall be provided on the front panel for purposes of synchronism of remote frequency indication and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±15 mc within the range at an ambient temperature of 20° C.

3.4.8.9 Local Oscillator and Crystal Mixer - The local oscillator shall be a coaxial cavity type using a reflex klystron tube. Its frequency shall be greater than the r-f pre-selector resonant frequency (the indicated receiver frequency) by the amount of the first i-f amplifier frequency (160 mc). The mixer shall use a suitable crystal. The input level to the mixer from the local oscillator shall be of such magnitude as to maintain the crystal current as indicated by the remote control metering circuit, within the range from 0.25 to 6 ma over the complete range of the tuner.

3.4.8.10 Sector-Sweep Adjustment - During the SECTOR-SWEEP operation, it shall be possible to adjust the frequency interval from the remote control, C-654A/APR-9B, to any desired interval within the tuning range down to a minimum interval of no greater than 250 mc.

3.4.9 R-F Tuner TN-131C/APR-9 (7050 - 10750 mc) - This r-f tuner shall fulfill the following requirements:

3.4.9.1 Form Factor and Mounting - This unit shall be housed in a size A1-D case in accordance with Specification MIL-C-172, and shall be supplied with an MT-A1-D mounting base fitted with vibration isolators.

3.4.9.2 Contents - This unit shall contain the following major items:

(1) Three-Cavity r-f preselector

(2) Reflex-Klystron local oscillator

(3) Crystal mixer

(4) I-F preamplifier

(5) Local oscillator and i-f preamplifier filament transformer

(6) Local oscillator repeller voltage divider circuit

(7) Sector-Sweep control potentiometer

(8) Synchro generator for remote frequency indication

(9) Motor-Drive tuning mechanism

(10) Frequency indicator counter

(11) Radar blanking input circuit receptacle

(a) Each r-f tuner shall be provided with a connection from the radar blanking input receptacle through an appropriate d-c blocking capacitor to the reflector terminal of the local oscillator tube. The circuit shall permit blanking of the tuner when a pulse voltage in the range from 40 to 215 volts peak-to-peak is applied to the radar blanking input receptacle. The applied blanking pulse is to have a duration from 6 to 15 usec and a repetition rate of

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

from 20 to 20000 pulses per second. Spurious responses resulting from the reflector modulation of the local oscillator reflector shall be at least 6- db below the desired response.

3.4.9.3 Weight with Mounting - The weight of this unit shall not be greater than 23 pounds, with the mounting base included.

3.4.9.4 Electrical Connections - Electrical connections shall be made to the front panel and shall consist of the following:

Power and control input receptacle
R-F input receptacle
I-F preamplifier output receptacle
Radar blanking input receptacle

3.4.9.5 Controls - A local tuning control for bench servicing, with provisions for disengaging motor drive (paragraph 3.3.6), shall be located on the front panel.

3.4.9.6 Limit Switches - For purposes of equipment protection, limit switches shall be provided at each end of the tuning range to remove the exciting voltages from the tuning motor. Operation of the limit switch at the low-frequency end of the tuning range shall simultaneously apply a voltage at the ID-226B/APR-9 indicator, which will illuminate the synchronizing light (paragraph 3.4.13.3.4). It shall be possible to adjust this limit switch so that it will operate at the indicated r-f tuner frequency of 7050 + 4/-0 mc. At the high-frequency end of the tuning range, such adjustments shall be possible at the indicated r-f tuner frequency of 10750 ±8 mc.

3.4.9.7 Tuning Rate - The tuning rate under condition of SECTOR-SWEEP operation shall be such that the time required to cover the complete frequency range of the r-f tuner is between the limits of 60 and 100 seconds at 20° C and in less than 155 seconds at -55° C. This tuning rate shall be maintained for operation with MANUAL switch setting in the remote control C-654A/APR-9B. After removal of tuning motor excitation, the receiver tuning motion shall stop within a maximum interval of 4 mc at any frequency in the range of the r-f tuner.

3.4.9.8 Frequency Indicator Counter - A frequency indicator counter shall be provided on the front panel for purposes of synchronism of remote frequency indication and for bench servicing. This frequency counter shall indicate a received signal frequency to an accuracy of at least ±15 mc within the tuning range at an ambient temperature of 20° C.

3.4.9.9 Local Oscillator and Crystal Mixer - The local oscillator shall be a coaxial cavity type using a reflex klystron tube. Its frequency shall be less than the r-f preselector resonant frequency (the indicated receiver frequency) by the amount of the preamplifier center frequency (160 mc). The mixer shall employ a suitable crystal. The input level to the mixer from the local oscillator shall be of such magnitude as to maintain the crystal current, as indicated by the remote control metering circuit, between the limits of 0.15 and 6 ma within the range of the tuner.

3.4.9.10 Sector Sweep Adjustment - During SECTOR-SWEEP operation it shall be possible to adjust the frequency interval from the C-654A/APR-9B remote control to any desired interval within the tuning range down to a minimum of no greater than 250 mc.

3.4.10 Mixer Amplifier CV-124/APR-13 - The mixer amplifier shall fulfill the following requirements:

3.4.10.1 Function - The mixer amplifier shall receive signals over a 4-mc bandwidth centered at 36.7 mc in the TN-200 from the i-f preamplifier in r-f tuners TN-178/APR-13 and TN-179/APR-13, the mixer amplifier receiver's signals over a 5-mc bandwidth centered at 60 mc. It provides an audio output for aural reception, a wide band of narrow-band video output from which the characteristics of received signals can be determined, and a panoramic output which is used in ID-226B/APR-9 indicator for panoramic presentation of the i-f bandpass and an i-f output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.10.2 Form Factor and Mounting - The unit shall be housed in a size A1-D case and mounted with a size MT-A1-D vibration mount, MT-606A/APR-9, in accordance with Specification MIL-C-172.

3.4.10.3 Contents of Unit - The unit shall contain the following major items:

(1) 36.7- to 60-mc i-f converter

(2) Main i-f amplifier and second detector

(3) Narrow-Band i-f amplifier and detector

(4) Wide-Band, narrow-band, and panoramic video amplifiers

(5) Sweep oscillator and panoramic mixer

(6) Horizontal sweep generator

(7) Beat frequency oscillator

(8) Fixed oscillator

(9) Audio amplifier and pulse stretcher

(10) AGC circuits

(11) I-F output circuit

(12) Relay for wide-narrow operation

(13) Filament transformer

3.4.10.3.1 36.7- to 60-Mc I-F Converter - The 36.7- to 60-mc i-f converter shall consist of a 23.3-mc crystal-controlled oscillator and a balanced mixer having gain and selectivity characteristics so as to satisfy the requirements of paragraph 3.4.10.3.2.

3.4.10.3.2 Main I-F Amplifier and Second Detector - The main i-f amplifier and second detector shall have a center frequency of 60 mc, and with the 36.7-mc i-f converter included, shall have voltage amplification and selectivity characteristics necessary to satisfy the following requirements at both 36.7 and 60 mc. This shall apply under conditions of 0-db i-f attenuation and AGC off.

3.4.10.3.2.1 I-F Response at 36.7 MC - The bandwidth to the extreme 2-db point shall not be greater than 2 mc. The apparent power gain shall be no less than 52.5 db as measured with a 93-ohm signal source and at a second detector level of 0.6 ma dc. The expression "apparent power gain" is used here because the input power is a continuous-wave RMS signal, and the output is d-c power. Therefore, the apparent power gain is the ratio of d-c power output at the second detector load resistor to the c-w signal power input expressed in db.

3.4.10.3.2.2 I-F Response at 60 Mc - The bandwidth to the extreme 2-db point, using maximum gain, as the reference level, shall be at least 6.4 mc and shall be centered at 60 ±0.5 mc. Gain variations between the extreme 2-db point shall not be greater than 2 db. The apparent power gain shall be no less than 64.5 db as measured with a 93-ohm signal source and at a second detector level of 0.6 ma dc.

3.4.10.3.3 Narrow-Band I-F Amplifier and Detector - The narrow-band i-f amplifier shall have a center frequency of 11 mc. Its bandwidth shall be 0.25 ±0.05 mc as measured at the 3-db points, and shall be 0.6 ±0.1 mc as measured at the 20-db points. With the requirements of paragraphs 3.4.10.3.4 (1) and (2) satisfied, voltage amplification through the narrow-band i-f amplifier and detector, at maximum narrow-band i-f gain control setting, shall be such that the average peak noise voltage appearing at either video output receptacle is of the same order of magnitude (within 2-to-1) on both wide and narrow bandwidth operation. The range of the narrow-band i-f gain control shall be such that the pulse output as measured at the video panel receptacle can be adjusted to equal magnitudes on both wide and narrow bandwidth operation for a pulsed signal of 10-usec duration and pulse repetition frequency of 1000 cycles applied to the main i-f amplifier.

Source: https://assist.dla.mil -- Downloaded: 20260-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.10.3.4 Video Amplifiers -

(1) Wide-Band Video Amplifier -

(a) The wide-band video amplifier shall have an amplification and phase shift characteristic such that the peak voltage shall not decrease more than 3 db as the pulsewidth of a pulsed 60-mc r-f signal applied to the main i-f amplifier is decreased from 25 to 0.3 usec at a pulse repetition frequency of 5,000 cycles. The output shall be measured at either video panel receptacle with a high impedance device such as an oscilloscope capable of measuring the pulses defined above. In addition, the pulsewidth as measured at the 6-db point at the video panel receptacle shall not differ from that of the applied r-f pulse by more than ±20 percent as the pulsewidth is varied as above. This requirement shall apply with an output pulse level of 0.5 volt peak.

(b) The low-frequency 3-db point of the amplification characteristic with respect to the gain in the flat portion of the response shall be 100 cps or less as measuredat either video receptacle. The input signal shall be sine-wave modulated 30 percent. In addition, the voltage amplification shall be such that at a second detector level of 0.6 ma dc, an input signal sine-wave modulated 30 percent with 1000 cps shall produce a peak-to-peak output voltage of at least 0.8 volt. Pulse polarity at the video panel receptacles shall be positive, and shall have a peak output of at least 5 volts without overloading in the main i-f amplifier or the wide-band video amplifier.

(c) The wide-band amplifier output shall appear at both video panel receptacles when the bandwidth switch in the C-654A/APR-9B control unit is set to its wide position.

NOTE: Realignment of Wide-Band I-F Amplifier - Realignment of the wide-band i-f amplifiers is permissible, upon replacement of tubes, to maintain the specified bandpass characteristics, provided suitable adjustments and alignment controls are supplied to insure that proper alignment can be made and that complete alignment procedure is outlined in the Handbook of Maintenance Instructions.

(2) Narrow -Band Video Amplifier -

(a) The narrow-band video amplifier shall have an amplification and phase shift characteristic such that the peak voltage output as measured in paragraph 3.4.10.3.4 (1) shall not decrease more than 3 db as the pulsewidth of a pulsed r-f signal is decreased from 25 to 4 usec at a pulse repetition frequency of 1000 cycles. In addition, the pulsewidth as measured at the 6-db point at the video panel receptacle shall not differ from that of the applied r-f pulse by more than ±10 percent as the pulsewidth is varied as above.

(b) The low-frequency 3-db point of the amplification characteristic with respect to the gain in the flat portion of the response shall be 100 cycles or less as measured at either video receptacle as in paragraph 3.4.10.3.4 (1).

(c) The video voltage amplification shall be at least 0.4.

(d) Pulse polarity at the video panel receptacle shall be positive and shall have a peak output of at least 5 volts without the occurrence of overloading in the main i-f and narrow i-f amplifiers, the panoramic mixer, or the narrow-band video amplifier. The narrow-band video amplifier output shall appear at both video panel receptacles when the bandwidth switch in C-654A/APR-9B control is set to its narrow position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

(3) Panoramic Video Amplifier -

(a) The panoramic video amplifier shall have an amplification characteristic so as to satisfy the requirements of paragraph 3.3.9. The voltage amplification of the panoramic video amplifier shall be at least 10 db as measured from the narrow-band detector load resistor to the output of the narrow-band panoramic video amplifier.

(b) This amplifier shall provide signal voltages to the ID-226B/APR-9 indicator for panoramic presentation of signals appearing in the i-f passband.

3.4.10.3.5 Swept Oscillator and Panoramic Mixer - The extreme frequency limits of the swept oscillator shall be 46.5 -0.25 mc and 51.5 + 0.25 mc. Adjustments shall be provided for setting these frequency limits. The conversion gain variation shall be maintained within 4 db for a given signal whose input level to the panoramic mixer is constant throughout a 5-mc frequency range arithmetically centered at 60 mc. Operation of this oscillator shall occur only with the bandwidth switch at its wide setting. The sweep rate of the swept oscillator shall be between 20 and 35 sweeps per second (10 to 17.5 cycles per second).

3.4.10.3.6 Horizontal Sweep Generator - Linearity of the horizontal sweep generator output, as measured at the panoramic output receptacle, shall be maintained within ±10 percent of the total frequency deviation from 46.5 to 51.5 mc (swept oscillator limits). The sweep generator output shall be at least 25 volts peak-to-peak as measured with a high impedance device. The fundamental frequency of the sweep generator voltage shall be identical to the sweep rate of the swept oscillator.

3.4.10.3.7 Beat Frequency Oscillator - The frequency of the beat frequency oscillator shall be 60 ±0.5 mc, and an adjustment shall be provided for setting this frequency.

3.4.10.3.8 Fixed Oscillator - The frequency of the fixed oscillator shall be 49 ±0.25 mc, and an adjustment shall be provided for setting this frequency. Its input level to the panoramic mixer, with the swept oscillator inoperative, shall be of such magnitude as to produce the same conversion gain, within ±4 db, as that obtained with the swept oscillator operating alone at the same frequency. This oscillator shall operate with the bandwidth switch in the C-654A/APR-9 control unit at its narrow setting. In addition, a switch shall be contained in the control unit to provide a choice of fixed oscillator operation or non-operation with the bandwidth switch at its wide setting. With the bandwidth switch in the wide position and the fixed oscillator on, a pulsed r-f signal tuned through the main i-f amplifier passband shall produce additional responses across the entire horizontal sweep of the ID-226B/APR-9 indicator display when the true response is at 60 mc.

3.4.10.3.9 Audio Amplifier and Pulse Stretcher - The audio amplifier, including the pulse stretcher, shall have an amplification characteristic which is not more than 3 db below maximum response between 200 and 10,000 cps as measured at either audio output receptacle into a 600-ohm resistive load and below the overload level of the amplifier. The amplifier and pulse stretcher, when working into a 600-ohm resistive load, shall satisfy the following requirements:

(1) The voltage amplitude variation, with an input of 10-usec duration and 0.7-volt peak amplitude, shall be no more than 0.4 db with a variation of pulse repetition frequency from 10,000 to 50 pps.

(2) With a peak pulse input of 0.7 volt and a pulse repetition frequency of 5000 pps, the output pulse amplitude shall be at least 26 volts peak and shall measure at least 30 usec in width at the half-voltage point with a 10-usec pulse in.

(3) With a constant pulse repetition frequency of 5000 pps and a peak pulse input of 0.7 volt, the peak pulse output shall vary no more than 17 db as the pulsewidth is varied from 10 to 0.1 usec.

(4) With an input pulse of 5000 pps, a peak amplitude of 0.7 volt and a pulsewidth of 0.5 usec, the output pulse shall measure at least 20 usec at the half-voltage point.

3.4.10.3.10 AGC Circuits - The automatic gain control circuit shall be delayed, amplifier type, operating on voltages derived from the rectified audio-frequency components of detected signals. Its characteristics shall be such as to maintain the i-f and/or video output levels below the overload level for pulsed r-f signals up to 200 mv applied to the input of the r-f tuner. The pulsed r-f signal shall have a duration of 10

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the currentversion before use.

usec and a pulse repetition frequency of 200 cycles. The AGC characteristic shall be realized by controlling the gain of such stages in the main i-f amplifier and preamplifier as required, with the exception that no control may be applied to the first tube of the preamplifier.

(1) Circuits shall be provided for the selection of either 2-sec or 0.1-sec AGC time constant. A switch shall be provided in the C-654A/APR-9B control for the selection of either time constant. When the AGC is inoperative (AGC switch off), the AGC output shall be grounded.

(2) The range of the AGC bias control shall be such that the video output can be adjusted from just below the normal overload level for pulsed signals down to a level no greater than 1.0 volt peak.

3.4.10.3.11 I-F Output Circuit - A portion of the output voltage from the main i-f amplifier shall be made available at the i-f output panel receptacle. The voltage gain from the main i-f amplifier input receptacle to the i-f output receptacle at 60 mc shall be no less than 40 db as measured with a 93-ohm signal source, with the i-f output receptacle termniated in a 93-ohm load. Moreover, gain variation over a 5-mc band centered at 60 mc shall not be greater than 3 db.

3.4.10.4 Weight - The weight of the unit with mounting shall not be greater than 16 pounds.

3.4.10.5 Electrical Connections - Electrical connections shall be made to the front panel of this unit and shall consist of the following:

	Connecting Plug
Power and control input receptacle	P-514
I-F input receptacle	P-501
Two video output receptacles	P-510 and P-511
Two audio output receptacles	P-512 and P-516
Panoramic output receptacle	P-515
Second detector d-c current receptacle	P-513
I-F output receptacle	P-517

3.4.10.6 Controls - Controls of the second i-f gain and AGC delay bias shall be provided as screw-driver adjustments, accessible from the front panel.

3.4.11 Mixer-Amplifier CV-43C/APR-9 - The mixer-amplifier shall fulfill the following requirements:

3.4.11.1 Function - The mixer-amplifier shall receive signals over a 20-mc bandwidth centered at 160 mc. It shall provide an audio output for aural reception, a wide-band or narrow-band video output from which the characteristics of received signals can be determined, a panoramic output which is used in the ID-226B/APR-9 indicator for panoramic presentation of the i-f bandpass, and an i-f output.

3.4.11.2 Form Factor and Mounting - The unit shall be housed in a size A1-D case and mounted with a size MT-A1-D vibration mount, MT-606A/APR-9, in accordance with Specification MIL-C-172.

3.4.11.3 Contents of Unit - The unit shall contain the following:

(1) Main i-f amplifier and second detector

(2) Narrow-Band i-f amplifier and detector

(3) Wide-Band, narrow-band, and panoramic video amplifiers

(4) Sweep oscillator and panoramic mixer

(5) Horizontal sweep generator

(6) Fixed oscillator

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the currentversion before use.

(7) Beat frequency oscillator

(8) Audio amplifier and pulse stretcher

(9) AGC circuits

(10) I-F output circuit

(11) Relays for wide-narrow operation and horizontal sweep reversal

(12) Filament transformer

3.4.11.3.1 Main I-F Amplifier and Second Detector - The main i-f amplifier shall have a center frequency of 160 ±0.5 mc. Its bandwidth to the extreme 2.5-db point shall be at least 20 mc wide, using maximum gain as the reference level. Gain variations between the extreme 2.5-db point shall not be greater than 2.5 db. The apparent power gain shall be no less than 58.5 db as measured with a 93-ohm signal source and at a second detector level of 0.2 ma dc. This shall apply under conditions of 0 db i-f attenuation and AGC off.

3.4.11.3.2 Narrow-Band I-F Amplifier and Detector - The narrow-band i-f amplifier shall have a center frequency of 39 mc. Its bandwidth shall be between 0.60 and 0.85 mc as measured at the 3-db point, and shall be 2.2 ±0.25 mc as measured at the 20-db point. With the requirements of paragraphs 3.4.10.3.4 (1) and 3.4.10.3.4 (2) satisfied, voltage amplification through the narrow-band i-f amplifier and detector, (maximum narrow-band i-f gain control setting) shall be such that the average peak noise voltage appearing at either video output receptacle is of the same order of magnitude (within 8-to-1) on both wide and narrow bandwidth operation. The range of the narrow-band i-f gain control shall be such that the pulse output, as measured at the video panel receptacle, can be adjusted to equal magnitudes on both wide and narrow bandwidth operation for a pulsed signal of 10-usec duration and a pulse repetition frequency of 1000 cycles applied to the main i-f amplifier.

3.4.11.3.3 Video Amplifiers -

(1) Wide-Band Video Amplifier -

(a) The wide-band video amplifier shall have an amplification and phase shift characteristic such that the peak voltage shall not decrease more than 3 db as the pulsewidth of a pulsed 160-mc r-f signal applied to the main i-f amplifier is decreased from 10 to 0.2 usec at a pulse repetition frequency of 5,000 cycles. The output shall be measured at either video panel receptacle with a high impedance device, such as an oscilloscope, capable of measuring the pulses defined above. In addition, the pulsewidth as measured at the 6-db point at the video panel receptacle shall not differ from that of the applied r-f pulse by more than ±10 percent as the pulsewidth is varied above.

(b) The low-frequency 3-db point of the amplification characteristic with respect to the gain in the flat portion of the response shall be 100 cps or less as measured at either video receptacle. The input signal shall be sine-waved modulated 30 percent. In addition, the voltage amplification shall be such that at a second detector level of 0.6 ma dc, input signal sine-wave modulated 30 percent with 1000 cps shall produce a peak-to-peak output voltage of at least 0.7 volt.

(c) Pulse polarity at the video panel receptacles shall be positive and shall have a peak output of at least 5 volts without overloading in the main i-f amplifier or in the wide-band video amplifier.

(d) The wide-band video amplifier output shall appear at both video panel receptacles when the bandwidth switch in the C-654A/APR-9B receiver control is set to its wide position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

(2) Narrow-Band Video Amplifier -

(a) The narrow-band video amplifier shall have an amplification and phase shift characteristic such that the peak voltage output as measured in paragraph 3.4.10.3.4 (1) shall not decrease more than 3 db as the pulsewidth of a pulsed r-f signal is decreased from 10 to 2 usec at a pulse repetition frequency of 1000 cycles. In addition, the pulsewidth as measured at the 6-db point at the video panel receptacle shall not differ from that of the applied r-f pulse by more than ±10 percent as the pulsewidth is varied as above. The low-frequency 3-db point of the amplification characteristic, with respect to the gain in the flat portion of the response, shall be 100 cycles or less as measured at either video receptacle as in paragraph 3.4.10.3.4 (1). The video voltage amplification shall be between 0.3 and 0.5.

(b) Pulse polarity at the video panel receptacle shall be positive, and shall have a peak output of at least 5 volts without overloading in the main i-f and narrow-band i-f amplifiers, the panoramic mixer, or the narrow-band video amplifier.

(c) The narrow-band video amplifier output shall appear at both video panel receptacles when the bandwidth switch in the C-654A/APR-9B receiver control is set to its narrow position.

(3) Panoramic Video Amplifier - The panoramic video amplifier shall bave an amplification characteristic so as to satisfy the requirements of paragraph 3.3.9. The voltage amplification of the panoramic video amplifier shall be at least 10 db as measured from the narrow band detector load resistor to the output of the panoramic video amplifier. The amplifier shall provide signal voltages to the ID-226B/APR-9 indicator for panoramic presentation of signals appearing in the i-f bandpass.

3.4.11.3.4 Swept Oscillator and Panoramic Mixer - The extreme frequency limits of the swept oscillator shall be 111, + 0/-0.5 mc, and 131 + 0.5/-0 mc. Adjustments shall be provided for setting these frequency limits. The conversion gain variation shall be maintained within 4 db for a given signal whose input level to the panoramic mixer is constant throughout a 20-mc frequency range arithmetically centered at 160 mc. Operation of this oscillator shall occur only with the bandwidth switch at its wide setting. The sweep rate of the swept oscillator shall be between 20 and 35 sweeps per second (10 to 17.5 cps).

3.4.11.3.5 Horizontal Sweep Generator - Linearity of the horizontal sweep generator output as measured at the panoramic output receptacle shall be maintained within ±10 percent of the total frequency deviation from 111 to 131 mc (swept oscillator limits). The sweep generator output shall be at least 25 volts peak-to-peak as measured with a high impedance device. The fundamental frequency of the sweep generator voltage shall be identical to the sweep rate of the swept oscillator.

3.4.11.3.6 Fixed Oscillator - The frequency of the fixed oscillator shall be 121 ±0.5 mc, and an adjustment shall be provided for setting this frequency. The input level to the panoramic mixer, with the swept oscillator inoperative, shall be of such magnitude as to produce the same conversion gain, within ±4 db, as that obtained with the swept oscillator operating alone at the same frequency. The oscillator shall operate with the bandwidth switch in the C-654A/APR-9B receiver control at its narrow setting. In addition, a switch shall be contained in the control unit to provide a choice of fixed oscillator operation or non-operation with the bandwidth switch at its wide setting. With the bandwidth switch in the wide position and the fixed ocsillator on, a pulsed r-f signal tuned through the main i-f amplifier bandpass shall produce additional responses across the entire horizontal sweep of the ID-226B/APR-9 indicator display when the true response is at 160 mc.

3.4.11.3.7 Beat Frequency Oscillator - The beat frequency oscillator shall be 160 ±0.5 mc, and an adjustment shall be provided for setting this frequency. The input level to the main i-f amplifier shall be of such magnitude that, with the i-f attenuation control set to a value which reduces the no-signal rectified current to less than 0.06 ma, the increase in this current caused by turning on the beat frequency oscillator shall be at least 0.1 ma.

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.11.3.8 Audio Amplifier and Pulse Stretcher - The audio amplifier, without the pulse stretcher, shall have an amplification characteristic which is not more than 3 db below maximum response between 200 and 10,000 cps as measured at either audio output receptacle into a 600-ohm resistive load and below the overload level of the amplifier. The amplifier and pulse stretcher, when working into a 600-ohm resistive load, shall satisfy the following requirements:

(1) The voltage variation, with an input pulse of 10-usec duration and 0.7-volt peak amplitude, shall be no more than 4 db with a variation of pulse repetition frequency of 10,000 to 50 pps.

(2) With a 10-usec pulse, a peak pulse input of 0.7 volt, and pulse repetition frequency of 5000 pps, the output pulse amplitude shall be at least 26 volts peak and shall measure at least 32 usec in width at the half-voltage point.

(3) With a constant pulse repetition frequency of 5000 pps and a peak pulse input of 0.7 volt, the peak pulse output shall vary no more than 17 db as the pulsewidth is varied from 10 to 0.1 usec.

(4) With an input pulse of 5000 pps, a peak amplitude of 0.7 volt, and a pulsewidth of 0.1 usec, the output pulse shall measure at least 20 usec at the half-voltage point.

3.4.11.3.9 AGC Circuits - The automatic gain control circuit shall have a delayed and amplified type circuit, operating on voltages derived from the rectified audio-frequency components of detected signals. Its characteristics shall be such as to maintain the i-f and/or video output levels below the overload level for pulsed r-f signals up ot 200 mv applied to the input of the r-f tuner. The pulsed r-f signal shall have a duration of 10 usec and a pulse repetition frequency of 2000 pps. The AGC characteristic shall be realized by controlling the gain of such stages in the main i-f amplifier and preamplifier as required, with the exception that no control may be applied to the first tube of the preamplifier.

(1) Circuits shall be provided for the selection of either 2-sec or 0.1-sec AGC time constant. A switch shall be provided in the C-654A/APR-9B control unit for the selection of either time constant. In addition, when the AGC is inoperative (AGC switch off), the AGC output shall be grounded.

(2) The range of the AGC bias control shall be such that the video output can be adjusted from just below the normal overload level for pulsed signals down to a level no greater than 1.0-volt peak.

3.4.11.3.10 I-F Output Circuit - A portion of the output voltage from the main i-f amplifier shall be made available at the i-f output panel receptacle. The voltage gain from the main i-f amplifier input receptacle to the i-f output receptacle at 160 mc shall be no less than 36 db as measured with a 93-ohm signal source and with the i-f output receptacle terminated in a 93-ohm detector load. Moreover, gain variations over a 20-mc band, centered at 160 mc, shall not be greater than 3 db.

3.4.11.4 Weight - The weight of the unit with mounting shall not be greater than 16 pounds.

3.4.11.5 Electrical Connections - Electrical connections shall be made to the front panel of this unit and shall consist of the following:

	Mating Plug
Power and control input receptacle	P-514
I-F input receptacle	P-501
Two video output receptacles	P-510 and P-511
Two audio output receptacles	P-512 and P-516
Panoramic output receptacle	P-515
Second detector d-c current receptacle	P-513
I-F output receptacle	P-517

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.11.6 Controls - Controls for the second i-f gain and AGC delay bias shall be provided as screwdriver adjustments accessible from the front panel.

3.4.12 Control, Radar Set C-654A/APR-9B - The radar set control shall be in accordance with Specification MIL-C-6781 for type I controls, and in addition, shall fulfill the following requirements:

3.4.12.1 Function - This unit shall provide the control and metering functions required for remote operation of the AN/ALR-5, In addition, this unit shall control the sector scan of the receiver by stopping the scanning when a signal is received within the pass-band of the receiver. The automatic signal finder (ASF) section shall operate after a maximum of 25 pulses has been received. The ASF shall operate on pulses of not more than 50 mv, independent of repetition rate between the limits of 50 and 1000 pps and pulsewidths of 0.1 and 10 usec. The circuits shall provide satisfactory control on pulse signals having a peak output voltage no greater than twice the average peak voltage noise.

3.4.12.2 Form Factor and Mounting - The unit shall conform to the applicable requirements of Specification MIL-C-6781 for type I control panels.

3.4.12.3 Contents of Unit - The unit shall contain the following:

(1) All necessary switches and controls for remote operation of the receiver, including provision for remote setting of the sector-sweep limits.

(2) A meter with switching provisions for measuring a-c line voltage and mixer crystal current. The meter switch shall be a spring-loaded type, normally connecting the meter in the oscillator injection position.

(3) An ASF circuitry (with suitable controls and indicators), to stop the sector sweep scanning when a signal is received.

3.4.12.4 Weight - The weight of the unit shall not be greater than 6.5 pounds.

3.4.12.5 Electrical Connections - Electrical connections shall be made to the rear of the unit in accordance with Specification MIL-C-6781, except that the following receptacles shall be provided:

	Connecting Plug
Control output	P-801
Power input	P-802
Video input	P-1303
ASF auxiliary control	P-1304

3.4.12.6 Controls - All controls shall be placed on the front panel and shall operate freely according to good engineering practice. The panel controls are as follows:

(1) MAN TUNE, HI-LO frequency tuning control

(2) SECTOR-SWEEP-MAN sector-sweep-manual tune selector switch

(3) HI LIMIT and LO LIMIT sector-sweep limits

(4) IF ATTN DB i-f attenuation control detent release

(5) VOL volume control

(6) POWER-OFF main power switch

(7) AGC-SHORT-OUT-AGC LONG automatic gain control switch

(8) ASF-BFO-OUT-RESET beat frequency oscillator switch and ASF switch

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the currentversion before use.

(9) FO-OUT frequency oscillator switch

(10) BANDWIDTH WIDE-NARROW bandwidth switch

(11) BAND band switch

(12) LINE-XTAL meter switch

(13) ADJ level control for ASF

(14) ADVANCE sweep advance switch

3.4.12.6.1 I-F Attenuation Control - The i-f attenuation control shall be a continuously variable potentiometer with positive detent action. A detent release knob shall be provided to permit operation without the detent if desired. The detent steps shall be equally spaced and calibrated in relative db attenuation as follows: 0, 6, 12, 18, 24, 30, 36, 42, 48, 54, 64, 79, and 100. Relative calibration accuracy of these points shall be within ±10 percent in db. An indexing control shall be provided to permit adjustment of the "grass height" to the desired reference level on the ID-226B/APR-9 indicator display at 0-db attenuation setting.

3.4.12.7 Plastic Lighting Plate - The front panel of this unit shall be equipped with a plastic lighting plate meeting the requirements of Specification MIL-P-7788.

3.4.12.8 Panel Illumination - Panel lights shall be provided for illumination of the sector-sweep frequency scale, tuner selector switch scale, i-f attenuation control scale, meter and panel engraving, as specified in Specification MIL-P-7788. Provision shall be made for remote adjustment of the panel lights to the desired brilliance.

3.4.12.9 Indicators - Indicators shall be provided on the front panel of this unit as follows:

3.4.12.9.1 Meter (to indicate crystal current and a-c line voltage) - When the BAND switch is set to any one of the 50 to 100-mc tuner positions and the LINE - XTAL switch is in XTAL position, full-scale deflection of the meter shall represent a current of 1 ma ±10 percent in the metered circuit. When the BAND switch is set to any one of the AN/APR-9 tuner positions and the LINE - XTAL switch is in XTAL position, full-scale deflection of the meter shall represent a current of 5 ma ±10 percent in the metered circuit. Under the latter conditions, the meter shall introduce not more than 30 ohms resistance into the crystal circuit. Line voltage indication shall be direct reading and shall have an accuracy of ±10 percent at 115 volts RMS over a temperature range of -54° C to + 55° C. This meter shall be sealed, and gasket-sealed meter is permissible.

3.4.12.9.2 Sector-Sweep Frequency Scale (to indicate sweep frequency limits) - The sector-sweep frequency scale shall operate with the BAND switch so that operation of the receiver, with an r-f tuner corresponding to the illuminated sector-sweep frequency scale, will present r-f signals on the ID-226B/APR-9 indicator that progress across the oscilloscope screen from left to right as the receiver is tuned lower in frequency and right to left as the receiver is tuned higher in frequency as viewed facing the oscilloscope screen. In addition to the five frequency scales required for 50- to 1000-mc tuners, the four frequency scales required by the 1000- to 10750-mc tuners shall be available by turning the BAND selector knob. The sector-sweep HI and LO limit pointers shall indicate the frequency limits over the complete frequency scale to within an accuracy of ±10 percent. Adjustment of the sector-sweep limits shall be possible to a minimum sweep range of not greater than 10 percent of the indicated frequency range of any r-f tuner. As the scale is changed, the band switch shall provide separate contacts for operation of the remote switching unit for changing r-f tuning heads.

3.4.12.10 Neon Light Assembly (to indicate video level) - The LEVEL lamp shall monitor the incoming video and noise signal level to the ASF section.

3.4.12.10.1 Indicator Light (to give visual indication that scanning motor is stopped) - The indicator lamp located in the ADVANCE switch shall light when the ASF section has been triggered by intercepted signal.

3.4.12.11 ASF Circuits - The automatic signal finder circuits shall consist of the following:

(1) A video amplifier to amplify the signal from the mixer amplifier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

(2) A monostable multivibrator to shape the video pulses.

(3) An integrator to discriminate against random noise pulses.

(4) A d-c cathode follower to reduce the source impedance of the integrated voltage.

(5) A bistable multivibrator to operate a relay when the integrator reaches a predetermined threshold voltage.

(6) A relay to stop the scanning motor and provide remote control contacts.

3.4.13 Indicator ID-226B/APR-9 - The indicator shall fulfill the following requirements:

3.4.13.1 Function - This unit shall perform the function of amplification and panoramic display of received signals and indication of tuner frequency.

3.4.13.2 Form Factor and Mounting - The unit shall be housed in a size S-2 case and mounted with a size MT-S-2 vibration mount, MT-607/APR-9, in accordance with Specification MIL-C-172.

3.4.13.3 Contents - The unit shall contain the following major items:

(1) Horizontal deflection amplifier

(2) Vertical deflection amplifier

(3) Pulse stretcher and baseline stablizer (for vertical deflection amplifier)

(4) Remote frequency indicating selsyn and synchronizing circuits

(5) Calibrated oscilloscope screen

3.4.13.3.1 Horizontal deflection amplifier - The horizontal deflection amplifier shall have a phase and amplitude characteristic which will, in conjunction with the vertical deflection amplifier, fulfill the following retrace requirement: The lateral difference in signal trace on the oscilloscope screen for the two directions of horizontal sweep shall be no greater than 1/8 inch for c-w signal reception with the sweep generator operating normally (paragraph 3.4.11.3.5). This requirement shall be satisfied when a 1-inch high amplitude test signal is applied on the horizontal trace to a trace length of 2-3/4 inches. In addition, the horizontal sweep deflection versus characteristic presented on the oscilloscope screen shall be linear to at least ±10 percent of the total frequency deviation over its 20-mc range.

3.4.13.3.2 Vertical Deflection Amplifier - The vertical deflection amplifier without the pulse stretcher and baseline stablizer, shall have an amplification characteristic that is not more than 3 db below maximum response between 100 and 20000 cps as measured on the oscilloscope screen. Voltage amplification of the vertical deflection amplifier measured from the panoramic input receptacle shall be at least 30 db.

3.4.13.3.3 Pulse Stretcher and Baseline Stabilizer - The baseline stablizer circuit shall maintain the position of the baseline on the oscilloscope screen substantially constant during intervals of high-level signal reception. It shall also remove the effect of responses that produce deflections below the baseline. A pulse stretcher shall be included to provide improved visibility and better amplification of narrow pulses.

3.4.13.3.4 Remote Frequency Indicating Selsyn and Synchronizing Circuits - The synchronizing light shall illuminate with operation of the limit switch at the low-frequency end of any r-f tuner. This shall permit alignment of the remote frequency counter dial, at the indicator unit, to the indicated frequency of an r-f tuner to an accuracy of at least ±5 mc. This accuracy shall be maintained throughout the tuning range of any r-f tuner.

3.4.13.3.5 Calibrated Oscilloscope Screen - A screen shall be provided for the panoramic oscilloscope, and it shall have calibrations to assist in reading the field strength of received signals. This screen shall have a zero index which shall be a horizontal line against which the "grass height marks" above this representing signal heights of + 6 db and + 12 db.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.13.4 Weight with Mounting - The weight of the unit shall not be greater than 12 pounds.

3.4.13.5 Electrical Connections - Electrical connections shall be made to the rear of the case and shall consist of the following:

	Mating Plug
Power and control input receptacle	P-601
Panoramic input receptacle	P-602

3.4.13.6 Controls - The following controls shall be located on the front panel:

(1) V-CENT vertical trace position control

(2) H-CENT horizontal trace position control

(3) FOCUS focus control

(4) INTENSITY intensity control

(5) INT MOD intensity modulation control

(6) H-GAIN horizontal amplifier gain control

3.4.13.7 Voltage Amplification - Voltage amplification, with the pulse stretcher and baseline stablizer circuits included, shall be such, that with the receiver gain set to produce a maximum 0.15 ma, no signal rectified current in the first i-f second detector load, receiver noise will produce at least 1/8 inch and not more than 1-3/4 inches of "grass" on the oscilloscope screen (as measured to the average of the noise peaks) with the bandwidth switch in the control unit at either its WIDE or NARROW setting. A variation in amplitude of the "grass" in the WIDE position shall be less than 2-to-1 across the horizontal sweep.

3.4.13.8 Vertical Centering - It shall be possible to adjust the vertical position of the horizontal trace to any point within the center 1/3 of the cathode-ray tube screen by means of the vertical centering control.

3.4.13.9 Voltage Gain - The voltage gain of the horizontal amplifier as measured from the panoramic input receptacle to the oscilloscope screen shall be at least 27 db at full gain.

3.4.13.10 Indicators - The following indicators shall be provided on the front panel for panoramic presentation and remote frequency indication of received signals:

(1) Oscilloscope Screen with Calibrated Plastic Scale

(2) KILO-MEGACYCLES remote frequency counter dial

(3) Synchronizing light (for remote frequency counter)

3.4.14 Power Supply PP-2848/APR-9 - This unit shall fulfill the following requirements:

3.4.14.1 Function - This unit shall provide various d-c operating voltages required by the receiver and filament supply voltage for the ID-226B/APR-9.

3.4.14.2 Form Factor and Mounting - The unit shall be housed in a size B1-D1 case and mounted with a size MT-B1-D1 vibration mount, MT-605A/APR-9, in accordance with Specification MIL-C-172.

3.4.14.3 Contents - This unit shall contain necessary circuitry to develop the following:

(1) Minus 1610-volt regulated supply at 18 ma for the reflex klystron oscillators of r-f tuners TN-130/APR-9 and TN-131/APR-9.

(2) Minus 1100-volt regulated voltage supply at 4 ma for the oscilloscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

(3) Minus 650-volt regulated voltage supply at 35 ma for oscilloscope of r-f tuners.

(4) Plus 300-volt unregulated voltage supply at 90 ma for sector-sweep control, oscilloscope amplifiers, audio amplifiers, and AGC amplifier circuits.

(5) Plus 150-volt unregulated voltage supply at 80 ma for second mixer, second i-f amplifier, beat frequency oscillator, fixed oscillator, audio amplifier, sweep oscillator circuits, and oscilloscope horizontal sweep amplifier screens.

(6) Plus 105-volt regulated A-supply at 210 ma for i-f preamplifier, sector-sweep potentiometer, high and low sector-sweep potentiometers.

(7) Plus 105-volt regulated B-supply at 15 ma for wide-band video amplifier, AGC delay circuits, and first audio amplifier screen.

(8) Relays and control circuits associated with sector-sweep and interlock functions.

(9) Filament supply for the ID-226B/APR-9 indicator

3.4.14.3.1 Rectifiers and Voltage Regulator Circuits - All rectifiers shall be silicon-diffused junction-type. The -1610, -1100, -650, + 300 and + 150-volt supply rectifiers shall be encapsulated. All voltage regulator circuits shall consist of a saturable reactor and an associated transistor differential amplifier. All voltage reference elements used in the regulator circuits shall be temperature-compensated, diffused-junction-silicon type.

3.4.14.4 Weight - The weight of the unit shall not be greater than 28.5 pounds, with the mounting base included.

3.4.14.5 Electrical Connections - Electrical connections shall be made to the front panel and shall consist of the following:

	Mating Plug
Power and control output receptacle for r-f tuners	P-901
Power and control output receptacle for mixer amplifier	P-902
Power and control output receptacle for indicator	P-903
Power and control input receptacle for receiver control	P-905

3.4.14.6 Controls - One screwdriver-actuated switch to disable the -1610-volt supply shall be located on the front panel. All other adjustments shall be screwdriver adjustments not accessible from the front panel.

3.4.14.7 Electrical Requirements - The primary power requirements shall be in accordance with paragraph 3.2.12.

3.4.14.7.1 Ripple - Over the specified power supply frequency range (380-to-420 cps), the ripple voltage for the various voltage supplies operating under normal conditions with any r-f tuner shall not be greater than values listed herein.

Supply (volts)	Ripple Factor (%)
-1610 regulated	0.1
-1100 regulated	0.3
-650 regulated	0.1
+105 regulated (A)	0.025
+105 regulated (B)	0.1
+150 unregulated	0.2
+300 unregulated	0.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.14.7.2 Regulation and Adjustment - For line voltage variations of +5 percent or less at 115 volts RMS over the 380- to 420-cps range, the regulation of the regulated voltage supplies operating under normal conditions with any r-f tuner shall not be greater than the values listed herein.

Supply (volts)	Regulation (%)
-1610	±0.3
-1100	±2.0
-650	±1.0
+105 (A)	±1.0
+105 (B)	±1.0

A control shall be provided for adjustment of the -1610, -650, and +105 (A) regulated voltage supplies. Adjustments shall be possible within the limits listed herein.

Supply (volts)	Accuracy of Adjustment (volts)
-1610	±5
-650	±5
+105 (A)	±5

3.4.14.8 Interlock and Sector-Screen Control - The interlock relay shall energize only with application of a-c control power. When unenergized (with +26.5-volt primary power applied to the receiver), the interlock relay shall function so as to render the tuning motor inoperative under condition of either MANUAL MANUAL or SECTOR-SWEEP operation. Under these conditions, the synchro locking relay in the ID-226B/APR-9 indicator shall lock the remote frequency counter dial and synchro follower.

3.4.15 Switch Assembly SA-418A/ALR-8 - The switch assembly unit shall meet the following requirements:

3.4.15.1 Function - This unit shall provide the necessary switching facilities to permit remote selection of any one of four tuning ranges chosen from the AN/ALR-8 system.

3.4.15.2 Form Factor and Mounting - This unit shall be housed in a size B1-B case and mounted with a size MT-B1-B vibration mount, MT-1512/ALR-8, in accordance with Specification MIL-C-172.

3.4.15.3 Contents of Unit - This unit shall contain the following:

(1) One i-f coaxial switch

(2) One i-f coaxial relay

(3) High-Voltage circuit switch

(4) Power and control circuit switch

(5) Automatic switch mechanism

NOTE: The SA-418A/ALR-9 unit shall contain two rotary-type solenoids as distinct from the SA-418/ALR-8 which contains an automatic tuning mechanism and a solenoid. Mechanical, electrical, and functional interchangeability shall be maintained.

3.4.15.3.1 I-F Coaxial Switch - A 6-position coaxial switch shall be used to switch the i-f output of the tuners. The switch shall have the following characteristics:

(1) Operating life shall not be less than 20,000 operations.

(2) The r-f attenuation shall not be greater than 0.3 db below 200 mc as measured at the input and output jacks located on the panel of the switch assembly.

(3) R-F attenuation between circuits below 200 mc shall be greater than 60 db.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

3.4.15.3.2 I-F Coaxial Relay - A 2-position coaxial relay shall be used to switch the i-f inputs to the mixer amplifier. The relay shall be 28-volt d-c nominal solenoid-operated relay with a minimum pickup voltage of 20 volts, with the same characteristics as listed above for the coaxial switch.

3.4.15.3.3 High-Voltage Circuit Switch - High-voltage, hermetically-sealed, relays shall be used for switching the high-voltages for tuner operation. For simplified design reasons, the limitation of engaging TN-130/APR-9 and TN-131/APR-9 in only two of the four tuner input terminals to this unit is permissible. These relays shall have an operating life of at least 20,000 operations.

3.4.15.3.4 Power and Control Circuit Switch - Wafer switches shall be used to switch the necessary power and control circuits. In addition, wafer switches shall be used to return the tuners, which are connected but not in operation, to the reset position. These switches shall have an operating life of at least 20,000 operations.

3.4.15.3.5 Automatic Switch Mechanism - The automatic switch mechanism shall provide for all necessary switching of high-voltage, i-f, and power and control circuits in such a manner that switching of tuners will automatically select the necessary mixer amplifier for the tuning range. If a tuning range should be selected at the C-654A/APR-9 remote control unit, in which the associated tuner is not engaged at the switch assembly, none of the engaged tuners will operate. Provision shall be made for local setup.

3.4.15.4 Weight with Mounting - The weight of the unit including mount shall not be greater than 22 pounds.

3.4.15.5 Electrical Connections - The following electrical connections shall be provided at the front and/or rear of the unit:

(1) Power and control for remote and local switching

(2) Power and control for the four selected tuners

(3) Power and control for each of the two mixer amplifiers

(4) Power and control from power supply

(5) Power and control for direction-finder antenna switching

(6) Audio for intercommunication system

(7) Four i-f connections from the selected four tuners

(8) Two i-f connections for the two mixer amplifiers

(9) Panoramic output from each mixer amplifier

(10) Panoramic output connection to the indicator control unit

(11) Video output connection from each mixer amplifier

(12) Two video connections for auxiliary analyzers

3.4.15.6 Controls - The following controls and switches shall be accessible from the front panel

(1) Channel A tuner selector switch

(2) Channel B tuner selector switch

(3) Channel C tuner selector switch

(4) Channel D tuner selector switch

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the current version before use.

NOTE: The above controls are not required for SA-418A/ALR-8.

3.4.15.7 Indicators - The unit shall have two indicator dials located on the front panel. One dial shall indicate tuner nomenclature in agreement with the tuning range selected at the C-654A/APR-9B remote control unit. The second dial shall indicate which of the tuners is engaged, as well as an OFF position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

pulse voltage in the range of from 40 to 125 volts peak-to-peak is applied when a

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Reliability Quality Assurance Tests

(3) Special Tests

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the currentversion before use.

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration.. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the currentversion before use.

4.4.2 Reliability Quality Assurance Tests - Equipments which have successfully passed the individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094.

4.4.2.1 Test Level - The Test level shall be Level II (from Table I of Specification MIL-R-23094).

4.4.2.2 Accept-Reject Criteria - Figure 12 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.2.3 Detailed Test Procedures - The Detailed Test Procedures for the acceptance of the equipment under Specification MIL-R-23094 shall be prepared by the contractor and approved by the procuring activity. The test procedures shall include but not be limited to:

(1) The performance characteristics to be measured.

(2) The criteria for failure.

(3) Any preventive maintenance to be allowed at periodic intervals during the test.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the currentversion before use.

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:01Z
Check the source to verify that this is the currentversion before use.

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the currentversion before use.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages 27.5 ± 0.5 V DC and 115 ± 1.0 V AC, as applicable. These data are to be used in testing the equipment at installation points for compliance the minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:01Z
Check the source to verify that this is the currentversion before use.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:01Z
Check the source to verify that this is the current version before use.